INCH-POUND

MIL-H-15505K(NU)
31 May 89
SUPERSEDING
MIL-H-15505J(NU)
18 December 1984

MILITARY SPECIFICATION

HAT, SERVICE, WOMAN'S

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a service hat frame ensemble, consisting of a hat frame, band with an adjustable mount, and a detachable hat cover.

*1.2 Classification. The service hat frames, hat covers, and hat bands, procured as an ensemble or as hat covers separately, shall be in the following types and sizes as specified (see 6.2):

Type I - Hat Frame (Enlisted) - Polyester/Wool Tropical, Blue 3346

Type IV - Hat Cover - Cloth, Twill, Polyester/Cotton (Durable Press), White 3053

Schedule of sizes

20, 20-1/2, 21, 21-1/2, 22, 22-1/2, 23, 23-1/2, and 24

2. APPLICABLE DOCUMENTS

2.1 Government documents.

*2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8410

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

L-P-504	- Plastic Sheet and Film, Cellulose Acetate
V-T-276	- Thread, Cotton
NN-P-71	- Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
DDD-L-20	- Label: for Clothing, Equipage, and Tentage (General Use)
DDD-T-86	- Tape, Textile; Cotton, General Purpose (Unbleached, Bleached or Dyed)
PPP-B-566	- Boxes, Folding, Paperboard
PPP-B-636	- Boxes, Shipping, Fiberboard

MILITARY

MIL-C-368	- Cloth, Satin, Rayon and Cloth, Twill, Rayon
MIL-T-5038	- Tape, Textile and Webbing, Textile, Reinforcing Nylon
MIL-P-15011	- Pallets, Material Handling, Wood, Post Construction, 4-Way Entry
MIL-B-17757	- Boxes, Shipping, Fiberboard (Modular Sizes)
MIL-C-21115	- Cloth, Tropical: Wool, Polyester/Wool
MIL-C-24927	- Cloth, Twill, Polyester/Cotton (Durable Press)
MIL-T-43548	- Thread, Polyester Core: Cotton, Rayon, or Polyester - Covered
MIL-C-43920	- Cloth, Interlining, Cotton or Synthetic, and Nylon
MIL-C-87074	- Cloth, Twill, Acetate

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-2073-1	- DOD Materials Procedures for Development and Applications of Packaging Requirements
MIL-STD-2073-2	- Packaging Requirements Code
MS-27982	- Fasteners, Snap, Style 3 (Pronged Ring Head Type)

Source: https://assist.dla.mil -- Downloaded: 20260-18T06:17Z
Check the source to verify that this is the current version before use.

(Unless otherwise indicated, copies of federal and military specification, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. POSTAL MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

* 2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of documents which are DoD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the documents cited in the solicitation (see 6.2).

NATIONAL MOTOR FREIGHT TRAFFICE ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036-1401).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606-5808).

DEPARTMENT OF DEFENSE STANDARD COLOR CARD OF OFFICIAL STANDARDIZED SHADES FOR SEWING THREADS

(Application for copies should be addressed to the Color Association of the United States, Inc., 343 Lexington Avenue, New York, NY 10016-0927).

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY

T410 OM-83 - Grammage of Paper and Paperboard (Weight per Unit Area)
T411 OM-84 - Thickness of Caliber of Paper and Paperboard

(Application for copies should be addressed to the Technical Association of Pulp and Paper Industry, Technology Park/Atlanta, P.O. Box 10513, Atlanta, GA 30348-5113).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

*(Nongovernment standards and other publications are normally available from organizations which prepare or which distribute the documents. These documents may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Material.

3.3.1 Hat frame ensemble.

3.3.1.1 Basic material. The polyester/wool tropical material to be used for the brim, band and mount, and band streamer pieces on type I, class 1, hat frames shall conform to type III, class 3, of MIL-C-21115. The material shall be dyed to match blue shade 3346 and the piece number for the standard sample is 4-001369.

3.3.1.2 Interlining.

3.3.1.2.1 Cloth, interlining, (innershell). The interlining cloth used for the top and side (innershell) crown pieces and front visor and back brim reinforcement pieces shall conform to type II of MIL-C-43920 exept that dimensional stability shall be tested in accordance with test method 5590. The specimens shall be submerged for 1/2 hour only.

3.3.1.2.2 Cloth, interlining, (outershell). The cloth used for the side crown (outershell) interlining piece shall have 42 (±2) warp ends per inch of 30/2 count natural cotton and 35 (±2) picks per inch of a 1040 denier monofilament nylon. The color of the filling may be natural or black or a combination of the two. The cloth shall be coated with an insoluble finish. The unfinished weight of the cloth shall be a minimum of 7 ounces per square yard, and the finished weight shall be 7.5 to 8.0 ounces per square yard. Testing shall be as specified in 4.4.1.

*3.3.1.3 Lining. The material used as a lining for the hat crown shall conform to class 1 of MIL-C-368. Acetate twill conforming to MIL-C-87074 may be used as an alternate material. When the alternate lining is used, the yarn size may be 100 denier warp x 150 denier filling. The cloth shall be dyed to approximate the shade of the basic material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

3.3.1.4 Felt. The felt material used to pad the brim piece shall be a minimum of 30 percent wool with the balance viscose rayon. The felt material shall be dyed a suitable black shade, and shall conform to the physical requirements specified in Table I when tested as specified in 4.4.1.

Table I - Felt Requirements

Characteristic	Requirement	
	Minimum	Maximum
Weight, ounces per square yard	7.0	9.0
Thickness, inches	0.040	0.065

3.3.1.4.1 Non-fibrous materials. The starch and protein content, including chloroform-soluble and water-soluble material of the finished felt, shall be a minimum of 10 percent and maximum of 20 percent when tested as specified in 4.4.1.

3.3.1.5 Sheeting, plastic (interlining). The plastic sheeting used for the brim interlining and crown top (outershell) piece shall be an extruded sheet of ionomer resin (see 6.7) laminated with a sheet of nonwoven polyester on one side and a coating of a polyolefin based resin adhesive on the other side. Total thickness of the sheeting shall be 0.019 to 0.025 inch. The color of the sheet resin may be brown or any dark shade. The nonwoven polyester side shall be natural color. Testing shall be as specified in 4.4.1.

3.3.1.6 Sweatband. The sweatband shall be made of a plain weave grosgrain ribbon, consisting of a rayon warp and cotton filling with a woven edge. No loading agents shall be used. The sweatband ribbon shall be dyed a suitable black shade, approximating the shade of the basic material (see 3.3.1.1). The finished sweatband shall conform to the requirements specified above and Table II, when tested as specified in 4.4.1.

Table II - Sweatband Requirements

Characteristic	Requirement	
	Minimum	Maximum
Weight (ounces per linear yard)	0.15	0.20
Width (inches)	13/16	15/16
Yarns per inch		
Warp	122	-
Filling	30	-

3.3.1.6.1 pH. The pH value of the water extract of the finished sweatband ribbon shall be a minimum of 5.5 and maximum of 8.5, when tested as specified in 4.4.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

3.3.1.6.2 Colorfastness. The dyed finished sweatband ribbon shall show a minimum of "good" fastness to wet-dry cleaning, perspiration, and crocking when tested as specified in 4.4.1.

3.3.1.7 Thread. The thread for seaming and stitching the hat frames shall be either cotton conforming to V-T-276, types IB2, ticket no. 36, 4 ply; or IC1, or IC2, ticket numbers A, O, or OO, 3-ply as applicable; or a covered polyester thread conforming to ticket number 50 of MIL-T-43548. The thread used for all brim stitching operations, including attachment of the sweatband, shall be dyed to match Black AA, Cable No. 66043. All other stitching operations, excluding basting operations, shall be sewn with white or natural shade thread (White AH, Cable No. 66050 or Natural White AG, Cable No. 66049). The thread used for basting shall be at the option of the contractor.

3.3.1.7.1 Colorfastness. The dyed thread shall show fastness to light, wet-dry cleaning, and perspiration equal to or better than the standard sample. When a standard sample is not available, the dyed thread shall show a minimum of "good" fastness to light, wet-dry cleaning, and perspiration. Testing shall be performed as specified in V-T-276 or MIL-T-43548, as applicable.

3.3.1.8 Plastic lining. The plastic used to line the inside of top crown shall conform to type II of L-P-504 with a nominal thickness of 0.0075 inch.

* 3.3.1.9 Fasteners.

3.3.1.9.1 Snap fasteners. The snap fasteners used to secure the band to frame shall conform to MS 27982, finish b, except for the following stud and eyelet dimensions apply:

a. The overall height of the stud shall be 0.234 (±0.002) inch and the flange diameter shall be 0.335 (±0.002) inch.
b. The diameter of the stud shall be 0.156 (±0.002) inch.
c. The overall height of the eyelet post shall be 0.335 (±0.002) inch and the flange diameter shall be 0.375 (±0.002) inch.

3.3.1.9.2 Buttons. The flat black buttons to be attached at the side crown shall be Waterbury Co. part no. 01-00909, size 24, back 02, finish 10436 with a 1/4 (±1/64) inch screw shank or equal (see 4.4.1, 6.8, and 6.11).

3.3.1.9.3 Screw posts. The screw posts for attaching the side buttons shall be Waterbury Co. part no. 03-12052-00-00-10100 with a 1/4 (±1/64) inch shank or equal (see 4.4.1, 6.8, and 6.11).

3.3.1.9.4 Eyelets. The eyelets for the side band shall be E. P. Stimpson part no. GS 8-5 brass/black 100 or equal (see 4.4.1, 6.8, and 6.11).

* 3.3.1.9.5 Washers. The washers for securing the screw posts shall be either Waterbury Co. part no. 04-01023, size 00, back 00, gold finish 10100; or Electronic Fasteners Inc. part no. C12044-012-4, or equal (see 4.4.1, 6.8, and 6.11).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

3.3.1.10 Mount. The mount to be attached to the type I hat frame band, shall be constructed of the materials specified below and in accordance with Table III, operation 4. The finished design and configuration of the enlisted personnel mount shall conform to the mount guide template (see 3.5.3).

3.3.1.10.1 Stay material. The stay material used in the construction of hat band insignia mounts shall be a flexible impregnated paperboard containing 15 percent (minimum) inert mineral filler. The paperboard shall be made of a cellulosic fiber having a finished thickness of 0.025 (±0.001) inch and weight of 110.0 (±5 percent) pounds per 1,000 square feet. The color shall be black. Testing shall be performed as specified in 4.4.1.

3.3.1.10.2 Leatherette. The leatherette used to back hat insignia mounts shall be made of either a fabric or paper base conforming to the requirements of 3.3.1.10.2.1 and 3.3.1.10.2.2.

3.3.1.10.2.1 Fabric base. The leatherette shall be made of any suitable vinyl coated woven or knitted cotton material embossed with a simulated leather effect on one side only. The fabric shall have a minimum finished weight 14 ounces per square yard, and shall show a minimum of "good" fastness to crocking and water. Testing shall be performed as specified in 4.4.1.

3.3.1.10.2.2 Paper base. The leatherette shall consist of a polyvinyl chloride film bonded or laminated with a suitable adhesive to a latex impregnated (30-35 percent by weight) cellulose pulpboard sheet. The polyvinyl chloride film shall be black in color, embossed on one side only, with a finished thickness of 0.009 (±0.0005) inch. In addition, the polyvinyl chloride film shall show a minimum of "good" fastness to crocking and water. The latex impregnated pulpboard sheet shall have a finished thickness of 0.030 (±0.0015) inch. Testing shall be performed as specified in 4.4.1.

3.3.1.10.3 Mount loop tape. The mount loop tape shall be 1/2 inch wide, conforming to types I, II, or III of DDD-T-86, or MIL-T-5038. The color shall be black, approximating the shade of the polyester/wool tropical fabric hat band.

3.3.1.10.3.1 Colorfastness. The tape shall show "good" colorfastness to wet drycleaning, crocking, and laundering when tested as specified in 4.4.1.

3.3.1.11 Label. Each hat frame shall have a combination, identification, and size label conforming to type VI, class 4, of DDD-L-20 except the inscription shall be as shown below. The maximum size of the label shall be 1-3/4 inches high and 2-3/4 inches wide. The printing on the label shall show a minimum of "good" fastness to accelerated dry cleaning.

NAME:
SERVICE NO:
HAT, SERVICE, WOMAN'S
CONTRACT NO: DLA 100-00-0-0000 (EXAMPLE)
STOCK NO: 8410-00-000-0000 (EXAMPLE)
NAME OF CONTRACTOR:
NAME OF MANUFACTURER: (IF DIFFERENT FROM CONTRACTOR)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

3.3.2 Hat cover.

3.3.2.1 Basic material. The basic material for the removeable hat cover (top and side cover) shall be a polyester/cotton twill cloth conforming to MIL-C-24927, white shade 3053 and the piece number for the standard sample is 86246.

*3.3.2.2 Sweatband. The sweatband shall be 100% polyester grosgrain ribbon, 5/8 (+1/16) inch wide, and a white shade approximating the basic material. The finished grosgrain ribbon shall conform to the following requirements when tested as specified in 4.4.1 (see 6.4).

Characteristic	Requirement	
	Minimum	Maximum
Weight, ounces per linear yard	0.10	0.15
Yarns per inch:		
Warp	70	
Filling	30	

3.3.2.3 Thread. The thread for seaming and stitching the hat cover shall be either cotton conforming to type IC1, ticket number 0 or 00, 3-ply, of V-T-276, or a covered polyester thread conforming to size number 50 of MIL-T-43548. The color of the thread shall be white, approximating the shade of the basic material.

*3.3.2.4 Label. Each finished hat cover shall have a label conforming to type VI, class 2 of DDD-L-20, with the size to be expressed numerically. The printing on the label shall show fastness to laundering only. When the hat covers are procured separately, the applicable contract number shall also be indicated. In combination with the size label, each hat cover shall have a small commercial type care label with the following inscription and format:

SIZE: 22 1/2 (EXAMPLE)
STOCK NO: 8410-00-000-0000 (EXAMPLE)
CONTRACT NO: DLA-100-00-0-000 (EXAMPLE, WHEN PROCURED SEPARATELY)

Machine Wash Warm, Permanent Press Cycle
Tumble Dry Medium Setting, Remove Promptly
DO NOT USE CHLORINE BLEACH
or
NAVEDTRA 414-01-45-81, Formula II

3.4 Design.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

3.4.1 Hat frame ensemble. The hat frame ensemble shall consist of a hat frame with a stiffened oval shape crown, stiffened brim with rolled sides and straight front and back; a detachable hat cover; and detachable hat band and mount. The hat band shall have an adjustable mount (without insignia) and two streamers in back. The band shall be attached to the frame at sides with black buttons and screwposts. Eyelets shall be placed in the band and holes in the frame to accomodate the screwposts. The front and back of the band shall be held in place by snap fasteners. Each hat frame shall be furnished with two cloth hat covers as part of the ensemble. The hat covers shall be of corresponding hat frame size and will fit snugly without any evidence of distortion on either of the components (see figure 1).

3.4.2 Hat cover. The hat cover shall consist of a circular top piece, side piece, a grosgrain sweatband ribbon attached on the inside bottom edge, and four horizontal buttonholes constructed as specified in Table IV, operation 5 (see figure 2).

3.4.3 Figures. Figures 1 through 4 are furnished for information purposes only. When inconsistencies exist between the written specification and the figures, the written specification shall govern.

*3.5 Patterns and blocks. Minor modifications are permitted where necessary when using automatic equipment. These modifications shall not alter the dimensional, serviceability, or appearance requirements cited in the specification.

3.5.1 Hat frame. Standard patterns to be used in cutting the contractor's working patterns will be furnished by the Government (see 6.3). In addition, the Government will provide one set of blocks (brim, plug and crown) in the size range specified (see 1.2). The Government patterns, the contractor's working patterns and blocks shall not be altered in any way. The Government blocks are not to be used as working blocks, but are restricted only for use in establishing the contractor's working blocks. Unless otherwise specified herein, the seam allowances shall be as specified on the pattern parts. The government loaned blocks shall be returned in good condition upon completion of the contract.

3.5.2 Hat cover. Standard patterns to be used in cutting the contractor's working patterns will be furnished by the Government. In addition, the Government will provide one cover block in the size range specified. The Government patterns, the contractor's working patterns and cover blocks shall not be altered in any way. The Government blocks are not to be used as working blocks, but are restricted only for use in establishing the contractor's working blocks. The contractor shall assure that the hat covers properly fit the corresponding frame size without distortion to the frame or covers, and provide proper alignment of buttonholes with the snap fasteners and screwposts of the frame. When the hat cover is procured separately, the hat frames used to verify the fit of the covers will be furnished by the Government to the contractor. The Government loaned frames and blocks shall be returned in good condition upon completion of the contract or order.

3.5.3 Pattern parts. The component parts of the hat frame and hat cover shall be cut from materials as specified, and in accordance with the following pattern parts:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

Material	Pattern Parts	Cut Parts
	Type I - Hat Frame	
Polyester/wool tropical	Brim, top	1
	Brim, bottom	1
	Band	1
	Band streamer	1
Cloth, interlining (innershell)	Crown, top (innershell)	1
	Crown, side (innershell)	2
	Front visor reinforcement	1
	Back brim reinforcement	1
Felt	Felt brim interlining	1
Sheeting, Plastic	Plastic brim interlining	1
	Crown, top (outershell)	1
Cloth, interlining (outershell)	Crown, side (outershell)	2
Cloth, lining	Crown, top (lining)	1
	Crown, side (lining)	1
Plastic lining	Crown, top protector (plastic)	1
Templates	Brim marker	-
	Band shaper	-
	Mount guide (finished) 1/, 2/	-
	Type IV - Hat cover	
Cloth, cotton, double plain weave	Cover, top	1
	Cover, side	1

1/ The hole diameter shall be 1/8 inch with no tolerance on the diameter. The true location of the hole shall be 1/32 inch of true position.

2/ This pattern part shall be used as a guide for cutting all component materials of the adjustable mount (see Table III, operation 4).

3.6 Construction.

3.6.1 Stitches, seams, and stitchings. Stitch, seam, and stitching types specified in Tables III and IV shall conform to FED-STD-751. Wherever two or more methods for seam or stitch types are given for the same operation, any one of them may be used. Where stitch types 101 and 401 are used, the looper (underthread) shall be on the inside of the hat frame, hat band, or hat cover. Seam allowances shall be maintained with the seams properly sewn so that no raw edges, runoffs, pleats, or open seams will result. Thread tension shall be maintained so that there is no tight or loose stitching.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

3.6.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Tables III and IV.

3.6.3 Thread breaks and ends of seams. Ends of all seams and stitchings, when not caught in other seams or stitching, shall be backtacked not less than 1/4 inch. The ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Thread breaks shall be secured by stitching back of the break not less than 1/2 inch. Skipped stitches or thread breaks on 401 or 101 seaming, may be repaired using a 301 stitch type. Where stitch type 401 or 101 is permitted as an option, it shall not be used unless the ends are caught in other seams or stitching.

3.7 Manufacturing operations. The hat frames and hat covers shall be manufactured in accordance with operation requirements specified in Tables III and IV. Minor modifications are permitted where necessary when using automatic equipment. These modifications shall not alter the dimensional, serviceability, or appearance requirements cited in the specification. The contractor is not required to follow the exact sequence of operations listed, except for the following precaution:

NOTE: Care shall be exercised by the contractor throughout the brim operation of the hat frame (see Table III, operation 5) in order to assure consistent stability and retainment of its blocked shape under adverse weather conditions.

3.7.1 Shade and size marking. The polyester/wool tropical component parts of the applicable hat frame and all component parts of the hat covers, shall be marked, ticketed, or bundled to assure a uniform shade and size throughout the item. Interlining and lining materials need not be shade marked. Any method of shade and size marking may be used except:

1. Corrosive metal fastening devices.
2. Sew-on type markings on lining materials.
3. Adhesive type tickets which discolor or adhere to the material upon removal of tickets.

NOTE: The use of ink pad numbering machine, rubber stamp or pencil is allowed, provided the numbering does not show on the outside of the garment and wherever possible, is covered by the seam allowance.

3.7.2 Abbreviations in table of operations. The abbeviations used in tables III and IV are as follows:

Stch	-	Stitch
in	-	inch
Ndl	-	Needle
Bob	-	Bobbin
Lpr	-	Looper
Mchne	-	Machine
Brtck	-	Bartack
Comrcl	-	Commercial
smlr	-	similar
Btnhl	-	Buttonhole
incl	-	including
dbl	-	double
chnstch	-	chainstitch

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

TABLE III - CONSTRUCTION OF HAT FRAMES

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB LPR
1.	Cutting. a. The component parts of the hat frame ensemble shall be cut in strict accordance with the patterns furnished, which show size, shape, directional lines for cutting, and notches for proper assembly of all component parts. b. Brim parts shall be cut from one piece of polyester/wool material. The remaining polyester/wool component parts for the hat band, streamer, and mount may be cut from ends. Parts cut from ends shall approximate the shade of the base material. c. Cut mount loops of sufficient length to finish as specified in operation 11f.				
2.	Replacement of defective components. During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in Section 4, shall be removed from production and replaced with non-defective and properly matched components.				
3.	Shade and size marking. Brim parts cut from the polyester/wool tropical shall be marked, ticketed, or bundled in accordance with 3.7.1.				
4.	Make insignia mount. a. Cut the paperboard stay and leatherette materials to shape using the mount guide marker. Cut the polyester/wool material for the mount in accordance with the mount guide marker, adding sufficient fabric to turn over in accordance with operation 4b. b. Cement the impregnated paperboard stay to the polyester/wool material. Turn the edge of the cloth over the edge of the curved portion of the stay approximately 1/4 inch and cement the cloth edges down, forming a finished folded edge.				

Source: https://assist.dla.mil -- Downloaded: 2026l0-18T06:17Z
Check the source to verify that this is the currentversion before use.

TABLE III - CONSTRUCTION OF HAT FRAMES

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB	THREAD LPR
	c. The polyester/wool material shall not be folded over the straight cut edges of the stay piece.						
	d. Cement the leatherette piece to the back side of the cloth covered stay piece using a suitable adhesive.						
	e. Topstitch approximately 1/16 inch from the folded edge along outer curve of the cloth covered mount. Extend stitching down to bottom of mount (see figure 3).						
	f. A 1/8 inch perforation shall be made in the mount as indicated by the crosshair location on the mount guide marker.						
5.	Make brim (see 3.7).						
	a. The component parts of the brim assembly shall finish in the following order from the top of brim to the bottom of brim. (see figure 1).						
	One ply of the polyester/wool tropical as the brim top piece.						
	One ply of the felt brim interlining, abutted at center back.						
	One ply of the back brim reinforcement piece.						
	One ply of the front visor reinforcement piece.						
	One ply of the plastic brim interlining, abutted at back center.						
	One ply of the polyester/wool tropical as the brim bottom piece.						
	b. Place the adhesive side of the plastic brim interlining against the felt brim interlining according to notches on patterns. Baste stitch together, simultaneously inserting and positioning the front visor reinforcement piece and the back brim reinforcement piece in between the two plies according to notches on patterns.						

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

TABLE III - CONSTRUCTION OF HAT FRAMES

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB LPR
	OR					
	As an alternate, the back brim reinforcement piece may be positioned on the top side of the felt brim interlining.					
	c. Join ends of the brim top and brim bottom pieces face to face, and stitch 1/4 inch from the raw edge.	301 or 401	SSa-1	12-14	0/3	0/3 0/3
	d. Spread each seam open and press flat.					
	e. Superimpose the brim interlining assembly to the back side of the brim top piece, aligning the center of the butted seams. The brim top piece shall extend 1/4 inch beyond the outer felt edge and the inner edge shall be even.					
	f. Baste stitch around its outer and inner circumferences 1/2 inch from the raw edges, assuring that the center back seams are evenly aligned.					
	g. With face sides together, superimpose the brim top piece to brim bottom piece, aligning the centers of the butted seam.					
	h. Stitch around the outer circumference 1/16 inch from the raw edge of the felt, matching all notches and easing in the brim bottom piece uniformly and without puckers.	301	-	12-14	0/3	0/3
	NOTE: Stitching shall be through only the top and bottom brim pieces, and not through the interlining pieces.					
	i. Turn the brim bottom piece over the assembly and baste stitch around the inner circumference approximately 1/4 inch from felt edge.	301	-	6-8	0/3	0/3
	NOTE: The outer edge of the brim shall finish crisp and well defined, with no folds, pleats, or puckers.					

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

TABLE III - CONSTRUCTION OF HAT FRAMES

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB LPR
	j. Spiral stitch the entire brim with one continuous row of stitching. The number of spirals shall be 20 (±1) and will be verified in the following way: The spiral stitch on the brim assembly (without crown and cover) shall be visually counted at the widest portion of the brim. When brim is attached to crown there shall be 17 (±1) visible spirals.	301	SSa-1	10 -14	0/3	0/3
	k. Turn the assembled brim so that when fitted into the block, the felt side of interlining assembly shall be closest to the block while blocking and on the top side of the brim when finished.					
	l. Block brim to proper shape, using contractor's supplied blocks that have been duplicated from the government furnished blocks. The brim shall be steamed until pliable and then fitted and shaped to the block. Care should be taken to properly center the brim in the block and to prevent the brim from shifting position during the shaping procedure. After shaping insert the block plug to ensure proper size. Additional steaming may be necessary to retain desired shape. Remove brim and plug from the block and place in drying oven until throughly dry. When dry remove plug from brim.					
	m. With brim marker template, baste stitch or chalk a guide line along inner edge of brim for subsequent assembly of crown.					
	n. The finished width of the brim in all hat frame sizes shall be in accordance with the dimensions specified in 3.8.1.					
*6.	Assemble crown.					
	a. With a bight of approximately 3/16 inch, overedge stitch the top crown outer shell piece.	503 504 or 505	EFd-1	8-10	0/3 0/3 0/3	0/3 00/3 00/3
	Note: There are two crown assemblies, inner and outer.					

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

TABLE III - CONSTRUCTION OF HAT FRAMES

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB LPR
	b. On each assembly, overlap each end of side crown piece over the other side piece approximately 1/2 inch, and single or double needle stitch. When a single row of stitching is used, the line of stitching shall be 1/16 to 1/8 inch from the raw edge. When a double row of stitching is used, the gauge shall be 3/16 to 1/4 inch with the outer row not less than 1/16 inch from the raw edge.	301 or 401	LSa-1 or LSa-2	8-10	0/3 0/3 36/4 A/3
	c. On each assembly, position the top crown piece over the side crown piece approximately 1/4 inch and stitch approximately 1/8 inch from the overedging. As an alternate, on the inner assembly the side crown piece may be positioned on the top crown piece. The side crown seams shall be centered at center front and center back.	301 or 401	LSa-1	8-10	0/3 0/3 or 36/4 A/3 0/3 00/3 or 36/4 A/3
	NOTE: Prior to blocking, the innershell crown assembly shall be inserted within the outershell assembly, with the raw edges positioned between the two crowns and swung slightly off center to prevent front and back seams of each crown from superimposing on one another.				
	d. Block the crown as follows:				
	(1). Position the entire crown assembly snugly over a corresponding size block.				
	(2). Fasten a tie cord securely around the base of the block so as to hold the crown taut, or trim the bottom edges of crown side pieces.				
	(3). Steam press blocked crowns thoroughly using a heated oven or iron. Guide marks shall be made for positioning the two snap fastener studs and two button screwposts just prior to removing the crown from the block (see operation 9).				
	(4). Upon removal of the crowns from the blocks, the crowns shall be sufficiently air dryed.				

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

TABLE III - CONSTRUCTION OF HAT FRAMES

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB LPR
7.	Assemble crown lining.					
	a. The crown lining shall consist of a two (2) ply top (crown top lining material and an underside of plastic sheeting) and a one-piece bias cut side crown lining piece.					
	b. Superimpose the ends of the side crown lining piece, and stitch approximately 1/4 inch from the raw edge.	301 or 401	SSa-1	8-10	0/3	0/3 0/3
	c. With the plastic top crown protector sandwiched in between the crown top lining and the side crown lining piece, stitch the assembly all around its circumference 3/8 (+1/16) inch from the raw edge, simultaneously taking up the fullness of the side crown lining piece by evenly pleating all around. Each pleat shall be approximately 3/8 inch in width. As an alternate, the side crown lining piece may be pre-pleated before sewing the assembly.	101 301 or 401	SSa-1	12-14	0/3	- 0/3 0/3
*8.	Attach lining to crown.					
	a. Position the lining assembly into the crown interior, distributing the fullness around its entire circumference and with the plastic top crown protector facing the inside of hat as worn.					
	b. Stitch the lining assembly to side crown pieces around the bottom edge, approximately 1/8 inch from the raw edge.	101 301 or 401	SSa-1	10-12	0/3	- 0/3 0/3
	c. Trim raw edges of excess lining material.					
	d. With bottom edge of lining and side crown pieces turned under at the tie cord indentation marks on outside side crown, stitch around crown 1/16 inch from raw edge.	101 301 or 401	SSa-1	10-12 0/3 0/3	0/3 0/3	- 0/3 00/3
	OR					
	Stitch the lining to side crown pieces around the bottom edge, approximately 5/8 inch from the raw edge.					
	e. Trim lining piece and side crown piece at fold line around bottom edge.					

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

TABLE III - CONSTRUCTION OF HAT FRAMES

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB LPR
	f. Overedge stitch the lining and side crown pieces around the bottom edge.	503 504 or 505	SSa-1	8-10	0/3 0/3	0/3 00/3
	g. Position crown over brim using the chalk or stitch guide marks inserted in operation 5.m, and stitch all around its circumference simultaneously inserting and catching the identification label in the line of stitching (see 3.3.1.11). The label shall finish to the inside of the hat with the top edge "Name" segment facing towards the front of the hat frame, with all of the printing clearly visible when the sweatband is turned back. To ensure proper appearance of finished hat, it is critical that the crown and brim be centered (front and back) before stitching together.	301 or 401	LSb-1 or LSa-1	12-14	0/3 0/3	0/3 00/3
	h. Press to shape. Care shall be taken not to form sharp lines along the brim upturn area.					
	i. The finished crown shall conform to the measurements specified in 3.8.1.					
*9.	Attach snap fastener studs, screw posts, and make side holes. a. Attach and secure the snap fastener studs to the face side of the side crown pieces according to guide marks indicated in operation 6d with the center of each stud to finish 1/4 (±1/8) inch from the bottom edge of the crown. The snap fastener studs shall be positioned as follows: 1. One stud at center front. 2. One stud positioned 1 (±1/8) inch to left of center back. b. To accomodate the button screw posts, make one 1/4 inch diameter hole centered on each side in accordance with the guide marks indicated in operation 6d. Attach and secure a washer and screw post to each center side.					

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

TABLE III - CONSTRUCTION OF HAT FRAMES

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB LPR
10.	Attach sweatband.					
	a. With the ends of the black sweatband overlapped and folded under approximately 1/2 inch, position selvage edge of sweatband 3/8 to 5/8 inch from inner raw edge of crown and stitch along selvage. The folded ends of sweatband shall be approximately 1-1/2 inches from either side of center back.	301 or 401	LSa-1	12-14	0/3	0/3 or 00/3
	b. Wet sponge and hot press sweatband to shape.					
11.	Assemble band.					
	a. Fold band approximately in half in lengthwise direction, with the underply extending approximately 1/4 inch. Double stitch folded side of band 1/8 - 3/16 inch gauge, with outer row not less than 1/16 inch from the folded edge.	301 or 401	OSf-2	12-14	0/3	0/3 or 0/3
	b. Fold under the 1/4 inch extension, catching the raw edge of the band in the fold. Double stitch turned edge of band 1/8 - 3/16 inch gauge with outer row not less than 1/16 inch from folded edge.	301 or 401	EFz-2	12-14	0/3	0/3 or 00/3
	c. Steam and hot press band using the curved band shaper template.					
	NOTE: Guide marks shall be made on the band for positioning the two snap fastener sockets (at front and back) and two eyelets (one on each side) to correspond with the crown of the frame.					
	d. Secure snap fastener sockets according to guide marks. The sockets shall be inserted through the inner ply only. Secure eyelets according to guide marks. The eyelets shall be through the completed band.					
	e. Stitch ends of band together approximately 3/16 inch from the raw edges. Press seam open and flat.	301 or 401	SSa-1	12-14	0/3	0/3 or 00/3

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

TABLE III - CONSTRUCTION OF HAT FRAMES

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB LPR
	f. Two mount loops shall be symmetrically positioned on inside of band, with the center of each loop 1 1/2 inch from center of front stud fastener (see figure 3). Turn under each end of mount loop 1/4 to 3/8 inch and stitch across the entire width of each end of the loop. The stitching shall be through all plies and superimposed upon the outer edge stitching of band. The finished opening shall measure 1 1/8 to 1 3/16 inches.	301	LSd-1	12-14	0/3	0/3
	g. The finished band shall measure 1-5/16 to 1-7/16 inches wide.					
12.	Assemble band streamer.					
	a. With face side together, fold streamer lengthwise and stitch 1/4 inch from cut edge along ends and side of streamer leaving an opening along the side for turning. Stitching at ends of streamer shall form points.	301 or 401	SSe-3(a)	12-14	0/3 0/3	0/3 00/3
	b. Trim points and turn streamer to finished position working out points. Double stitch along all edges of streamer, turning in the seam allowance opening. The stitching shall be 1/16 and 1/4 inch from the edges and the two rows shall not cross each other.	301	SSe-3(b)	12-14	0/3	0/3
	c. The band streamer shall be pressed and finish 8 (±1/4) inches long and 1 3/8 (±1/8) inches wide.					
13.	Attach streamer to band.					
	The assembled streamer shall be doubled over the band forming a point at center back with the overlapping halves of the streamer angled diagonally, with the top ply towards the left and the bottom ply towards the right sides of the hat as worn (see figure 1). Invisibly stitch the properly positioned streamer to the band.					
14.	Finishing.					
	a. Trim and remove all loose ends of thread.					
	NOTE: Care shall be taken when triming thread ends that damage to fabric does not result.					

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

TABLE III - CONSTRUCTION OF HAT FRAMES

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB LPR
	b. Remove all spots, stains, and shade tickets.				
	c. Prepare applicable type, class, and size hat frame with corresponding hat covers for packaging and packing (see Section 5).				

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

TABLE IV - CONSTRUCTION OF HAT COVERS

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
1.	Cutting.					
	a. The component parts of the hat cover shall be cut in strict accordance with the Government patterns which show size, shape, directional lines for cutting, and notches for proper assembly of the component parts.					
	b. All component parts shall be cut from one piece of material.					
	c. Cut the grosgrain ribbon 1/2 inch longer than the applicable hat size.					
2.	Replacement of defective components. During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in Section 4, shall be removed from production and replaced with non-defective and properly matched components.					
3.	Shade and size marking. All component parts shall be marked, ticketed, or bundled in accordance with 3.7.1.					
4.	Assemble hat cover.					
	a. With face sides together, stitch ends of side piece 3/16 to 1/4 inch from the raw edge. Spread seam open and press flat.	301 or 401	SSz-3(a)	10-12	0/3	0/3 or 0/3 00/3
	b. Double stitch 1/8 to 3/16 inch gauge, with the seam centered between the two rows of stitching.	301 or 401	SSz-3(b)	10-12	0/3	0/3 or 0/3 00/3
	c. With face sides together, join side piece to top piece stitching 3/16 to 1/4 inch from the raw edges. Spread seam open and press flat, with the side piece seam positioned at center back of top piece.	301 or 401	SSz-3(a)	10-12	0/3	0/3 or 0/3 00/3
	d. Double stitch 1/8 to 3/16 inch gauge, with the seam centered between the two rows of stitching.	301 or 401	SSz-3(b)	10-12	0/3	0/3 or 0/3 00/3
	e. Join ends of the sweatband approximately 1/4 inch from the raw edges.	301 or 401	SSa-1	10-12	0/3	0/3 or 0/3 00/3

Source: https://assist.dla.mil -- Downloaded: 20260-18T06:17Z
Check the source to verify that this is the current version before use.

TABLE IV - CONSTRUCTION OF HAT COVERS

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
	f. The joined ends of the sweatband may be spread open or turned to one side when the sweatband is attached to bottom open edge of side piece.					
	g. Fold under the bottom edge of side piece 3/16 to 5/16 inch, and position one selvage of sweatband a minimum of 1/16 inch from the inner folded edge. Stitch around the entire inner circumference near the folded edge.	301	SSt-2	10-12	0/3	0/3
	h. Stitch other edge of the sweatband to the side piece, simultaneously inserting and catching the length of one side of label in the closing seam (see 3.3.2.4). The label shall finish 1 (±1/2) inch from either side of center back seam of side piece, face side up, with the printing clearly visible.	301	-	10-12	0/3	0/3
5.	Insert buttonholes. Make four horizontal straight style buttonholes with a cut opening of 7/8 (±1/16) inch in sweatband. Position the center of each buttonhole 1/4 (± 1/16) inch from and parallel to the bottom edge of sweatband. All buttonholes shall be barred at each end and positioned as follows: 1. One at the center front. 2. One centered on each side. 3. One positioned 1 (±1/8) inch to left of center back seam. NOTE: Buttonhole location shall ensure that the hat frame's snap fastener posts shall freely penetrate the buttonholes without distortion to the cover, when attached to the hat frame and covered with band.	Button-hole		45 per button-hole	00/3	00/3
6.	Finishing and blocking. a. Turn the hat cover to its finished position, with the seams and label on the inside. b. Trim and remove all loose ends of thread. NOTE: Care shall be taken when triming thread ends that damage to fabric does not result.					

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

TABLE IV - CONSTRUCTION OF HAT COVERS

No.	Operation	Stch Type	Seam/ Stch type	Stch in	Thread Ndl	Thread Bob/ Lpr
	c. Remove all spots, stains, and shade tickets.					
	d. Each hat cover shall be fitted over the corresponding size blocks duplicated from the Government furnished blocks.					
	e. Insert each blocked hat cover into a heated steam oven for approximately 1 minute, with the temperature of the oven properly controlled.					
	f. The hat covers shall be completely air dried prior to removal from the blocks, and upon removal they shall be free from wrinkles.					

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

3.8 Finished measurements.

3.8.1 Hat frames. The sizes and measurements of the finished hat frames shall be in strict accordance with Table V.

Table V - Finished measurements

*Measurement 1/	SIZE 20	20-1/2	21	21-1/2	22	22-1/2	23	23-1/2	24
Brim Width:									
A Center front	2-1/2	2-1/2	2-1/2	2-1/2	2-1/2	2-1/2	2-1/2	2-1/2	2-1/2
B Center side	2-3/4	2-3/4	2-3/4	2-3/4	2-3/4	2-3/4	2-3/4	2-3/4	2-3/4
C Center back	1-7/8	1-7/8	1-7/8	1-7/8	1-7/8	1-7/8	1-7/8	1-7/8	1-7/8
Crown Top:									
D Center length	5-3/4	5-7/8	6	6-1/8	6-1/4	6-3/8	6-1/2	6-5/8	6-3/4
E Center width	5-1/8	5-1/4	5-3/8	5-1/2	5-5/8	5-3/4	5-7/8	6	6-1/8
Crown Height:									
F Center front	2-3/8	2-3/8	2-3/8	2-7/16	2-1/2	2-9/16	2-5/8	2-11/16	2-3/4
G Center side 2/	2-1/2	2-1/2	2-1/2	2-9/16	2-5/8	2-11/16	2-3/4	2-13/16	2-7/8
H Center back	2-5/8	2-5/8	2-5/8	2-11/16	2-3/4	2-13/16	2-7/8	2-15/16	3

1/ All measurements are in inches and a tolerance of ± 1/8 inch is permitted. Crown dimensions are taken with cover removed from frame.

2/ The difference between the two center side measurements shall not exceed 1/8 inch. If this condition occurs it shall be scored as a measurement defect.

A) Brim width: Center front - Measure brim depth at center front extending from bottom edge of crown to finished edge of brim. Take measurement at center front snap fastener post.

B) Brim width: Center sides - Measure brim in flat position 4 1/4 inches from front snap fastener post, from bottom edge of crown to finished edge of brim.

C) Brim width: Center back - Measure depth of brim at center back seam, from bottom edge of crown to finished edge of brim.

D) Crown top: Center length - Measure top of crown from center front edge to center back edge.

E) Crown top: Center width - Measure across top of crown from center side to center side edge.

F) Crown height: Center front - Measure height of crown from bottom edge of crown to top crown seam, adjacent to center front snap fastener post.

G) Crown height: Center sides - Measure sides 4 1/4 inches from center front snap fastener post, from bottom edge of crown to edge of top crown piece.

H) Crown height: Center back - Measure height of crown at center back seam, from bottom edge of crown to edge of top crown piece.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

*3.8.2 Hat covers. The sizes and measurements of the finished hat covers shall be in accordance with Table VI. All measurements and tolerances are expressed in inches. The hat covers shall smoothly fit the hat ensemble or when procured separately, the covers shall smoothly fit the government furnished frames.

Table VI - Finished Measurements

Measurement 1/	SIZE 20	20-1/2	21	21-1/2	22	22-1/2	23	23-1/2	24
Cover Top:									
I Center length	5-7/8	6	6-1/8	6-1/4	6-3/8	6-1/2	6-5/8	6-3/4	6-7/8
J Center width	5-1/4	5-3/8	5-1/2	5-5/8	5-3/4	5-7/8	6	6-1/8	6-1/4
K Cover Height:	2-5/8	2-5/8	2-5/8	2-3/4	2-3/4	2-3/4	2-3/4	2-7/8	2-7/8

1/ All measurements are in inches and a tolerance of ± 1/8 inch is permitted.

I) Center length - Measure top of cover from center front edge to center back seam.

J) Center width - Measure top of cover from center side to center side seam.

K) Cover height - Measure height of cover from bottom edge of cover to top seam at center front.

3.9 Workmanship. The finished hat frame ensembles and hat covers shall conform to the quality of product established by this specification. The occurence of defects shall not exceed the applicable point value specified in Table VIII.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.2 and 6.2), it shall be examined for the defects listed specified in 4.4.2.3 and 4.4.2.4.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated herein.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents. A certificate of compliance shall be submitted and will be acceptable for the hat cover sweatband requirements (see 3.3.2.2 and 6.4), for the fastener hardwear requirements (see 3.3.1.9.1 through 3.3.1.9.5), and for the hat box requirements (see 5.1.1.1). In addition, components and materials listed in Table VII shall be tested for the characteristics specified and in accordance with the referenced test methods of FED-STD-191 whenever applicable. All test reports shall contain the individual values utilized in expressing the final result. The sample size shall be as shown below, and the lot shall be unacceptable if one or more units fail to meet any requirements specified.

Lot Size	Sample Size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

VII - Component Testing

Component (Lot size unit)	Characteristic	Requirement Paragraph	Test Method
Felt (yds)	Mat'l identification		
	Fiber	3.3.1.4	1/
	Color	3.3.1.4	1/
	Weight	3.3.1.4	5041
	Thickness	3.3.1.4	5030
	Nonfibrous	3.3.1.4.1	2611 1/
Sweatband Ribbon, Black, Hat Frame (yds)	Mat'l Identification	3.3.1.6	1/
	Weave	3.3.1.6	Visual
	Weight	3.3.1.6	5041
	Color	3.3.1.6	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

Table VII - Component Testing (cont.)

Component	Characteristic	Requirement Paragraph	Test Method
	Width	3.3.1.6	5020
	Thread count (warp & filling)	3.3.1.6	5050
	pH	3.3.1.6.1	2811
	Colorfastness:		
	Wet dry cleaning	3.3.1.6.2	5622
	Perspiration	3.3.1.6.2	5680
	Crocking	3.3.1.6.2	5651
Plastic Sheeting Interlining (yds)	Mat'l Identification	3.3.1.5	1/
	Thickness	3.3.1.5	5030 1/
	Color	3.3.1.5	Visual
Cloth Interlining (Outershell)	Mat'l Identification	3.3.1.2.2	1/
	Thread per inch	3.3.1.2.2	5050 1/
	Weight	3.3.1.2.2	5041 1/
	Color	3.3.1.2.2	Visual
	Yarn size	3.3.1.2.2	1/
Paperboard (sheets or sq. ft.)	Mat'l Identification	3.3.1.10.1	1/
	Percent filler	3.3.1.10.1	1/
	Thickness	3.3.1.10.1	T411 1/ 2/
	Weight	3.3.1.10.1	T410 1/ 2/
Leatherette Fabric or Paper Base (yards)	Mat'l identification	3.3.1.10.2	1/
	Coating ident.	3.3.1.10.2	1/
	Weight	3.3.1.10.2	5041 1/
	Thickness	3.3.1.10.2	5030 1/
	Colorfastness:		
	Crocking	3.3.1.10.2	5651 1/
	Water	3.3.1.10.2	5630 1/
Tape, Cotton or Nylon (yds)	Colorfastness:		
	Wet drycleaning	3.3.1.10.3.1	5622
	Crocking	3.3.1.10.3.1	5651
	Laundering	3.3.1.10.3.1	5614
Sweatband Ribbon, White, Hat Cover (yds)	Mat'l identification	3.3.2.2	1/
	Weight	3.3.2.2	5041
	Thread count	3.3.2.2	5050

1/ A certificate of compliance shall be submitted and will be acceptable for the characteristics specified.

2/ Testing shall be performed in accordance with procedures of the Technical Association of the Pulp and Paper Industry.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

4.4.1.1 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operations or assemblies are carried out as specified. The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated non-conformance.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 and 4.4.2.2.

4.4.2.1 Hat frame ensemble. The finished hat frame ensemble shall be packaged in the unit knock-down box (see 5.1.1.1), and shall be examined in accordance with 4.4.2.3 and 4.4.2.4. Sampling provisions shall be as indicated in accordance with Table VIII. The sample size shall be based on lot size. The sample unit shall be the entire contents within a hat box. The lot size shall be expressed in terms of the sample unit. The lot shall be unacceptable if:

1. The point value for 3 & 2 point defects exceeds the applicable maximum acceptable point value.

OR

2. The point value for total (3, 2, & 1) defects exceeds the applicable maximum acceptable point value.

4.4.2.1.1 Determination of outside/inside. The terms "outside" and "inside" as used throughout this examination, are defined as follows:

Outside: hat frame - Any part of hat frame that is visible at shoulder level height when positioned as worn. The hat frame shall be completely rotated.

Outside: hat band with attached mount and streamers - Any part of band, mount, or streamer that is visible at shoulder level height when positioned as worn. The band assembly shall be completely rotated.

Outside: hat cover - Any part of hat cover that will be visible when worn over the crown portion of hat frame. The hat cover shall be completely rotated.

Inside: hat frame - Any part of hat frame that is not visible at shoulder level height when positioned as worn. The hat frame shall be completely rotated.

Inside: hat band with attached mount and streamers - Any part of band, mount, or streamer that is not visible at shoulder level height when positioned as worn. The band assembly shall be completely rotated.

Inside: hat cover - Any part of hat cover that will not be visible when worn over the crown portion of hat frame. The hat cover shall be completely rotated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

lining shall consist of a two ... d. The crown li...

4.4.2.2 Hat covers (procured separately). The finished hat covers as packaged in 5.1.1.2, shall be examined in accordance with 4.4.2.3 and 4.4.2.4. The applicable inspection levels and AQL's shall be as indicated in 4.4.2.5. Sampling provisions shall be as indicated in accordance with Table VIII. The sample size shall be based on lot size. The sample unit shall be one finished hat cover. The lot size shall be expressed in terms of the sample unit. The lot shall be unacceptable if:

1. The point value for 3 & 2 point defects exceeds the applicable maximum acceptable point value.

OR

2. The point value for total (3, 2, and 1 point) defects exceeds the applicable maximum point value.

The Government will furnish hat frames in the appropriate sizes to be used during the examination for fit.

4.4.2.3 Visual examination. The finished hat frame ensemble or the finished hat cover (when procured separately) shall be examined for defects in design, material, construction, workmanship, marking, dimension, fit and classified in accordance with the list below. (The sample unit shall be one completely finished unit (hat frame ensemble or hat cover).

Classification of defects

Defect	Point Value
I Material Defects and Workmanship Damages	
a. Any hole, cut, tear, smash, or broken yarn	3
b. Needle chew, loose yarn, snagged yarn or thread, coarse yarn, visible mend, thin place, woven in waste, shade bar, dye streak, or area of no dye penetration.	
1. On outside; excluding the crown portion of hat frame (longest dimension in any direction):	
- up to 1/4 inch, inclusive	2
- more than 1/4 inch	3
2. On outside crown portion of hat frame (longest dimension in any direction):	
- up to 1/4 inch, inclusive	1
- more than 1/4 inch	2
3. On inside (longest dimension in any direction):	
- 1/4 inch up to and including 1/2 inch	1
- more than 1/2 inch	2

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

Defect	Point Value
c. Knot or slub on outside 1/	1

1/ Only a knot or slub which exceeds the limit shown on Sears Fabric Defect Scales (see 6.5), "C" or "4" as applicable for slubs and "B" for knots shall be scored.

II Shaded parts

Any part shaded or variation in shade within a part

1. On outside, excluding the crown portion of hat frame	2
2. On outside crown portion of hat frame	1
3. On inside	1

III Cleanness

a. Spot or stain

1. On outside; excluding the crown portion of hat frame (longest dimension in any direction):

- up to 1/4 inch, inclusive	1
- more than 1/4 inch	2

2. On outside crown portion of hat frame (longest dimension in any direction):

- more than 1/4 inch	1

3. On inside (longest dimension in any direction):

- 1/4 inch up to and including 1/2 inch	1
- more than 1/2 inch	2

b. Thread ends

Outside - more than 3 thread ends not trimmed to 1/4 inch or less	1
Inside - more than 5 thread ends not trimmed to 1/4 inch or less	2

c. One or more shade or size tickets not removed	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

Defect	Point Value
IV Blocking and Pressing (brim, crown and cover)	
a. Blocking and/or pressing omitted	3
b. Poorly blocked and/or pressed, i.e., evidence of gloss marks; shine; or wrinkles.	2
c. Asymmetrical blocking of brim	3
V Component and Assembly	
a. Any component part of operation omitted	3
b. Any component part not as specified	2
c. Any operation not performed as specified	2
d. Crown not properly positioned on brim causing misalignment of front notches and subsequent misalignment of studs	2
VI Seams and Stitchings	
a. Accuracy of seaming	
1. Seam irregular, pleated, twisted, puckered or wavy	1
2. End of stitching, when not caught in another seam or stitching, not finished as specified	1
3. End of a continuous line of stitching overlapped less than 1/2 inch	1
4. Stitching overlapping end of thread break less than 1/2 inch	1
5. Not specified color or varying shades of thread used on outside	2
6. Uneven spacing of spiral stitching	1
b. Gauge of stitching and seam allowance	
1. Seam allowance not as specified by 1/16 inch or more (score only when the condition exists or more than 1/2 the length of seam) irregular, not within the range specified.	2
or	
2. Stitching varies more than 1/16 inch when no range is specified (score only when the condition exists on more than 1/2 the length of seam).	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

Defect	Point Value
c. Open seam 1/	
1. On outside:	
- up to 1/4 inch, inclusive	1
- more than 1/4 inch to 1/2 inch inclusive	2
- more than 1/2 inch	3
2. On inside:	
- up to 1/4 inch, inclusive	1
- more than 1/4 inch to 1 inch inclusive	2
- more than 1 inch	3

1/ Two or more broken skipped stitches on joining seams shall constitute an open seam. On double stitched seams, a seam is considered open when one or both stitches of the seam is open.

d. Runoff, skipped or broken stitch:	
1. Joining seam - use "Open seam" classification	
2. Top or raised stitching:	
- 1/4 inch up to 1/2 inch inclusive	1
- more than 1/2 inch	2
e. Raw edge:	
1. On outside, excluding the overlapped front and back seams of crown on hat frame	
- up to 1/4 inch, inclusive	1
- more than 1/4 inch	2
f. Seam and stitch type	
1. Not specified seam or stitch type	2
2. Any line of stitching omitted	3
3. Any line of stitching not beginning or ending where specified (unless otherwise classified herein)	1
4. Use of 401 or 101 stitch type has looper thread on outside.	2

Source: https://assist.dla.mil -- Downloaded: 202610-18T06:17Z
Check the source to verify that this is the currentversion before use.

Defect	Point Value
g. Stitch tension:	
1. Loose tension resulting in a loose seam	2
2. Loose exposed loops of lower or top thread on edge or raised stitching.	2
3. Tight tension (stitches break when normal strain is applied in the direction of the seam or stitching)	3
h. Stitches per inch (score only when condition exists on more than 1/2 the length of seam):	
1. More than the maximum specified	1
2. Less than the minimum specified	
- by one or two stitches	1
- by three or more stitches	2
VII Crown lining, hat frame	
a. Twisted or loosely attached	2
b. Pleats not evenly distributed, evidenced by excessive fullness in an area or throughout	1
VIII Brim, hat frame	
a. Twisted or poorly shaped	3
b. Finished turned up side edges of the brim varies 1/8 inch or more in height or from the crown sides	3

NOTE: To determine this condition, place the hat on a flat surface. For the side brim height measurement, measure from the flat surface to the top edge of brim at maximum height of side, with the measuring device perpendicular to the flat surface. For the gap measurement (at maximum height of side brim), measure from the crown to the outer edge of the brim. These measurements should be taken in the same location on both sides.

c. Number of visible spiral stitching on brim not 17 (±1). Count visible spirals from outer edge of brim to inner edge of crown. Cap cover and band shall be removed for this inspection.

IX Sweatband, hat frame or hat cover as applicable	
a. Twisted or loosely attached	2
b. Sweatband exposed below bottom edge of hat cover by more than 1/16 inch	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

Defect	Point Value
c. Finished width of hat frame sweatband, less than 13/16 inch or more than 15/16 inch	1
d. Finished width of hat cover sweatband, less than 9/16 inch or more than 11/16 inch	1
X Band and streamer assembly, hat frame	
a. Twisted, puckered, or poorly shaped	2
b. Mount not conforming to the configuration of the mount guide template	3
c. Finished mount on band when attached to the hat frame and viewed from the center front distorted; crooked; or not in alignment with center front of brim	
1. Seriously affecting appearance	3
2. Not seriously affecting appearance	2
d. Construction of mount not as specified	3
e. Any evidence of cement visible on face side of mount	3
f. Thread ends visible at base of curve on mount	1
g. Perforation for screwpost hole in mount omitted, not as specified	3
h. Color of background cloth for face side of mount not as specified	2
i. Color of leatherette (mount backing) not as specified.	1
j. Band streamers not securely stitched or incorrectly positioned to band	2
k. Streamer ends rounded, not forming sharp points	2
l. Finished width of band or streamers less than 1-1/4 inches or more than 1-1/2 inches	1
m. Band with mount too tight when positioned on hat creating visible bulge	2
XI Snap fasteners	
a. Any stud portion of snap fastener omitted on crown portion of hat frame	3
b. Any stud portion of snap fastener misplaced or insecurely attached on crown portion of hat frame	2

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

Defect	Point Value
c. Any socket portion of snap fastener omitted on band assembly	3
d. Any socket portion of snap fastener misplaced or insecurely attached on band assembly	2
XII Buttonholes, hat cover	
a. Any omitted, incorrect style, uncut, or incomplete stitching	3
b. Broken stitch or two or more skipped stitches in two or more buttonholes	2
c. Tacking at ends omitted or not serving intended purpose	1
d. Stitches per buttonhole less than minimum specified:	
1. Two to four stitches	1
2. More than four stitches	2
e. Cut length less than 13/16 inch or more than 15/16 inch	2
f. Buttonhole mispositioned, not parallel with bottom of cover	1
XIII Combination label, hat frame	
a. Missing, incorrect, illegible	2
b. Misplaced or insecurely attached as specified	1
c. Label size larger than 1-3/4 inches high and 2-3/4 inches long	1
XIV Size/care label, hat cover	
a. Missing, incorrect, or illegible	2
b. Misplaced, insecurely or improperly attached	1
XV Eyelet, button, washer, and screwpost	
a. Any omitted	3
b. Any misplaced, loose, not positioned or attached as specified	2
c. Finish omitted or not as specified	2
d. Any broken or malformed	3
e. Any having a sharp or rough edge	2
f. Any clinched too tightly	3

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

4.4.2.4 Dimensional and fit examination (hat frame). The appropriate number of hat frames drawn from the hat boxes and determined from the table below, shall be examined for conformance to the dimensional requirements cited in Table V along with the fit of hat covers included within. When a hat frame measurement deviates from a dimension and tolerance specified, the hat frame shall be penalized one point. The hat cover, fit tested against the applicable size hat frame, and dimensional requirements cited in Table VI shall be scored for defects as noted below. Each size hat frame present in the lot shall be represented in the sample selected for these examinations. The lot shall be unacceptable if the total point value resulting from either one of these examinations exceeds the maximum acceptable point value for the lot (see 4.4.2.5).

4.4.2.5 Hat covers (procured separately). The appropriate number of hat covers, determined as specified in the table below, shall be fit tested against the dimensional requirements cited in Table VI and the applicable size hat frames furnished by the government and scored for defects noted below. When a hat cover measurement deviates from a dimension and tolerance specified, the hat cover shall be penalized one point. The lot shall be unacceptable if the total point value resulting from this examination exceeds the maximum acceptable point values. Each size hat cover present in the lot, should be represented in the sample selected.

Fit test of hat covers

Defect	Point Value
1. Cover size not compatible with frame size	1
2. Cover tight, causing the crown and or sides of frame to buckle	1
3. Cover loose, causing a loose fit or puckering condition	1
4. Buttonholes do not line up with snap fastener and screwpost studs on frame, causing twist or bulge.	1

Sampling provisions for dimensional and fit examinations

Lot Size	Sample size	Maximum acceptable point values
Up thru 500	8	0
501 thru 3,200	13	1
3,201 thru 35,000	20	2
35,001 and up	32	3

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

Table VIII - Sampling Provisions

	Lot size	Sample size 1/	Maximum acceptable point values 3 & 2 point	3, 2, & 1 point
Normal Inspection	Up thru 90	20	7 points	8 points
	91 thru 150	32	9 points	13 points
	151 thru 280	50	12 points	18 points
	281 thru 500	80	18 points	27 points
	501 thru 1200	125	26 points	39 points
	1201 thru 3200	200	38 points	58 points
	3201 thru 10,000	315	55 points	83 points
	10,001 thru 35,000	500	80 points	120 points
	35,001 and over	800	118 points	176 points
Tightened Inspection	Up thru 90	20	4 points	6 points
	91 thru 150	32	6 points	10 points
	151 thru 280	50	9 points	14 points
	281 thru 500	80	15 points	24 points
	501 thru 1200	125	22 points	34 points
	1201 thru 3200	200	32 points	55 points
	3201 thru 10,000	315	50 points	78 points
	10,001 thru 35,000	500	74 points	116 points
	35,001 and over	800	113 points	176 points
Reduced Inspection	Up thru 280	20	6-10 points 2/	13 points
	281 thru 500	32	9-14 points	18 points
	501 thru 1200	50	13-18 points	23 points
	1201 thru 3200	80	18-23 points	32 points
	3201 thru 10,000	125	26-31 points	44 points
	10,001 thru 35,000	200	38-43 points	62 points
	35,001 and over	315	55-60 points	88 points

1/ If sample size equals or exceeds lot or batch size, do 100 percent inspection.
2/ For reduced inspection, if the first value is exceeded but the second value has not been exceeded, accept the lot but return to normal inspection. The second value is the maximum acceptable point value.

4.4.2.6 Inspection procedures.

4.4.2.6.1 Initiation of inspection. Normal inspection shall be used at the start of inspection unless otherwise directed by the responsible procurement quality assurance element administering the contract.

4.4.2.6.2 Continuation of inspection. Normal, tightened, or reduced inspection shall continue unchanged on successive lots except where switching procedures in 4.4.2.6.3 require change.

4.4.2.6.3 Switching procedures.

4.4.2.6.3.1 Normal to tightened. When normal inspection is in effect, tightened inspection shall be instituted when two out of five consecutive lots have been rejected on original inspection (i.e. ignoring resubmitted lots for this procedure).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

4.4.2.6.3.2 Tightened to normal. When tightened inspection is in effect, normal inspection shall be instituted when five consecutive lots have been considered acceptable on original inspection.

4.4.2.6.3.3 Normal to reduced. When normal inspection is in effect, reduced inspection shall be instituted providing that all of the following conditions are satisfied:

a. The preceding 10 lots have been on normal inspection and none have been rejected on original inspection.

b. The total number of points for 3 and 2 point defects in the samples from the preceding 10 lots is equal to or less than sixty percent of the total maximum acceptable point values for 3 and 2 point defects from the preceding 10 lots.

c. The total number of points for 3, 2, and 1 point defects in the sample from the preceding 10 lots is equal to or less than seventy-five percent of the total maximum acceptable point values for 3, 2, and 1 point defects from the preceding 10 lots.

d. Production is at a steady rate.

e. Reduced inspection is considered desirable by the procurement quality assurance element administering the contract.

4.4.2.6.4 Reduced to normal. When reduced inspection is in effect, normal inspection may be instituted if any of the following occur on original inspection:

a. A lot is rejected.

b. A lot is considered acceptable but exceeds the applicable first value (see Table VIII, 2/).

c. Production becomes irregular or delayed.

d. Other conditions warrant that normal inspection be instituted.

4.4.3 Examination of packaging requirements. An examination shall be made to determine that packaging, packing, and marking comply with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be on the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 202610-18T06:17Z
Check the source to verify that this is the current version before use.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling, bulged or distorted container.
Content	Number of items per shipping container is more or less than required. Size shown on one or more items not as specified on shipping container.1/

1/ For this defect, one item from each shipping container in the sample shall be examined.

4.4.4 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized units loads in the end item inspected lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5 in accordance with MIL-STD-105.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of load not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Hat frame ensemble. Each finished type hat frame, packaged as an ensemble, shall have a type IV hat cover, and mount with band assembled and attached to the frame. A replacement cover shall be included with the ensemble and shall be packaged as follows:

Source: https://assist.dla.mil -- Downloaded: 202610-18T06:17Z
Check the source to verify that this is the current version before use.

- Each finished and assembled hat frame shall be wrapped with tissue paper around the center top and inserted snugly against the sides of the brim.

- Position the hat frame over a snug-fitting cylindrical paperboard support using additional tissue paper in the crown of the hat frame and around the entire hat, thus preventing any shifting within the box. The cylindrical paperboard support shall be 3 (+1/8) inches high, with the ends overlapped approximately 1/2 inch, and secured by either taping, stapling, or any other mechanical means.

- The additional hat cover shall be folded to form a three part oval, with the two side pieces overlapping each other beneath the top piece of the cover and shall be inserted into a clear polyethylene bag of 0.00125 inch (+ 25 percent) thickness. The polyethylene bag shall be formed with heat sealed seams that are straight, continuous and parallel to each other and the formed edges of the bag, and shall be a minimum of 1 inch longer than the hat cover. The bag closure shall be heat sealed, made as close as possible to the open end. A 1/4 inch diameter hole shall be made at one corner of each polyethylene bag to allow excess air to escape.

The individually packaged hat cover of corresponding hat frame size shall be placed into the following hat box, along with the hat frame ensemble:

- The square style knock down box shall conform to variety 1, group III, similar to style X (see Figure 4) of PPP-B-566 with a tuck n'lock bottom and a folding lock top with protruding handle to carry the box. The box shall have finished interior dimensions of approximately 10-3/4 inches (L), x 11-1/2 inches (D), x 6 inches (W) sized to accomodate the complete size range (20-24). The handle shall be on the 10-3/4 x 6 inches top. The box shall be marked with either a label or printed markings on the top side near the handle and wrapped in white paper.

5.1.1.2 Hat covers (when procured separately). Each type IV hat cover folded as specified in 5.1.1.1, shall be inserted into a snug-fitting, flat style, clear polyethylene film bag of 0.00125 inch (+ 25 percent) thickness. The polyethylene bag shall be formed with heat sealed seams that are straight, continuous and parallel to each other and the formed edges of the bag, and shall not be a minimum of 1 inch longer than the hat cover. The bag closure shall be heat sealed, made as close as possible to the open end. A 1/4 inch diameter hole shall be made at one corner of each polyethylene bag to allow excess air to escape.

5.1.2 Level C (Commercial). Hat frame ensembles furnished with replacement hat covers, or hat covers procured separately shall be preserved to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

5.2.1.1 Hat frame ensemble. Twenty-four (24) hat frame ensembles, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type SF, class weather-resistant, grade V2s, of PPP-B-636.

Inside dimensions of each container shall approximate 33 inches in length, 24 inches in width, and 24 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.3.

5.2.1.2 Hat covers (when procured separately). One hundred fifty (150) type IV hat covers of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to type SF, class weather-resistant, grade V2s, size 1, of MIL-B-17757. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper. Level A packages shall be packed uniformly throughout the shipping container. Closure requirements for the shipping container shall be as specified in the appendix of the box specification.

5.2.2 Level B

5.2.2.1 Hat frame ensemble. Twenty-four (24) hat frame ensembles packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety SW or SF), class domestic, grade 275, of PPP-B-636. Level A packages shall be packed flat, three in length, two in width, four in depth. Inside dimensions of each container shall approximate 33 inches in length, 24 inches in width, and 24 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.3.

5.2.2.2 Hat covers (when procured separately). One hundred fifty (150) type IV hat covers of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to type CF, class domestic, grade 275, size 1, of MIL-B-17757. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper. Level A packages shall be packed uniformly throughout the shipping container. Closure requirements for the shipping container shall be as specified in the appendix of the box specification.

5.2.3 Level C. Hat frame ensembles or type IV hat covers when procured separately, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall be in accordance with US Postal Service Manual, Uniform Freight Classification Rules, or National Motor Freight Classification Rules as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

* 5.3 Marking. In addition to any special marking required by the contract or order, unit packs, intermediate, or exterior containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

5.4 Palletization. When specified (see 6.2) item packed as specified shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C, K, and L or O or P. Pallet pattern shall be in accordance with the appendix of MIL-STD-147.

The pallet shall be 4-way, Type IV; Type V, Class 1, Size 2; or Type VIII, fabricated from wood group I, II, III, or IV, Grade A of NN-P-71, or 4-way, Style 1, Size A, Type I, Class 1 fabricated from wood groups specified of MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

* 5.5 MIL-STD-2073. When specified (see 6.2) Preservation and Packing shall be in accordance with MIL-STD-2073-1 and MIL-STD-2073-2.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hat frame ensembles with replacement hat covers are intended for use by female Navy personnel as part of non-utility uniforms.

*6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Size and type required (see 1.2)
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)
d. When first article sample is required (see 3.2). The item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quality, testing and approval of the first article.
e. Levels of packaging and packing required (see 5.1 and 5.2)
f. When palletization is required (see 5.4)
g. When all packaging data is to be found in MIL-STD-2073-1 and MIL-STD-2073-2

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids.

6.4 Sweatband, polyester grosgrain. A polyester grosgrain ribbon for the type IV hat cover, conforming to the requirements of 3.3.2.2 may be obtained from Olympic Narrow Fabrics Company, Inc., P.O. Box 631, Graham, NC 27253 or equal (see 6.11).

6.5 Sears Fabric Defect Scales are available from Sears, Roebuck and Company, Department 817, (ATTN: BSC 23-29), Sears Tower, Chicago, IL 60684.

*6.6 Hat box. A knock down hat box conforming to the requirements of 5.1 may be obtained from Foremost Corrugated Company, Inc., P.O. Box 2208, South Hackensack, NJ 07606. Other suppliers may obtain approval of their item by submitting it for approval to contracting officer.

6.7 Sheeting, plastic (interlining). Material meeting the requirements of 3.3.1.5 may be obtained from Bixby International Corp, 53 Warren Street, Newburyport, MA 01950 designated as XL-15-R-SC.

*6.8 Hardware. The screw shank button (3.3.1.9.2), screwpost (3.3.1.9.3), and washer (3.3.1.9.5) are available from the Waterbury Company, Inc., P.O. Box 1812, 835 South Plain Street, Waterbury, CT 06720. The eyelets (3.3.1.9.4) are available from E.P. Stimpson, 900 Sylvan Avenue, Bayport, NY 11705. An acceptable washer (3.3.1.9.5) is also available from Electronic Fasteners, Inc., 305 Winter Street, Waltham, MA 02154.

*6.9 Recycled materials. It is encouraged that recycled materials be used when practical as long as it meets the requirements of this specification.

*6.10 Additional white and khaki crowns are available as optional items and shall meet the requirements specified throughout the certification program.

6.11 Equal item. Prior to the use of an "or equal" item, the contractor shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval.

*6.12 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a courtesy only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

*6.13 Subject term (key word) listing.

Cover, detachable
Hat, combination
Hatband and streamer
Headwear, woman's dress

Custodian:

Navy - NU

Review Activity:

DLA - CT

Preparing Activity:

Navy - NU

Project 8410-0639

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

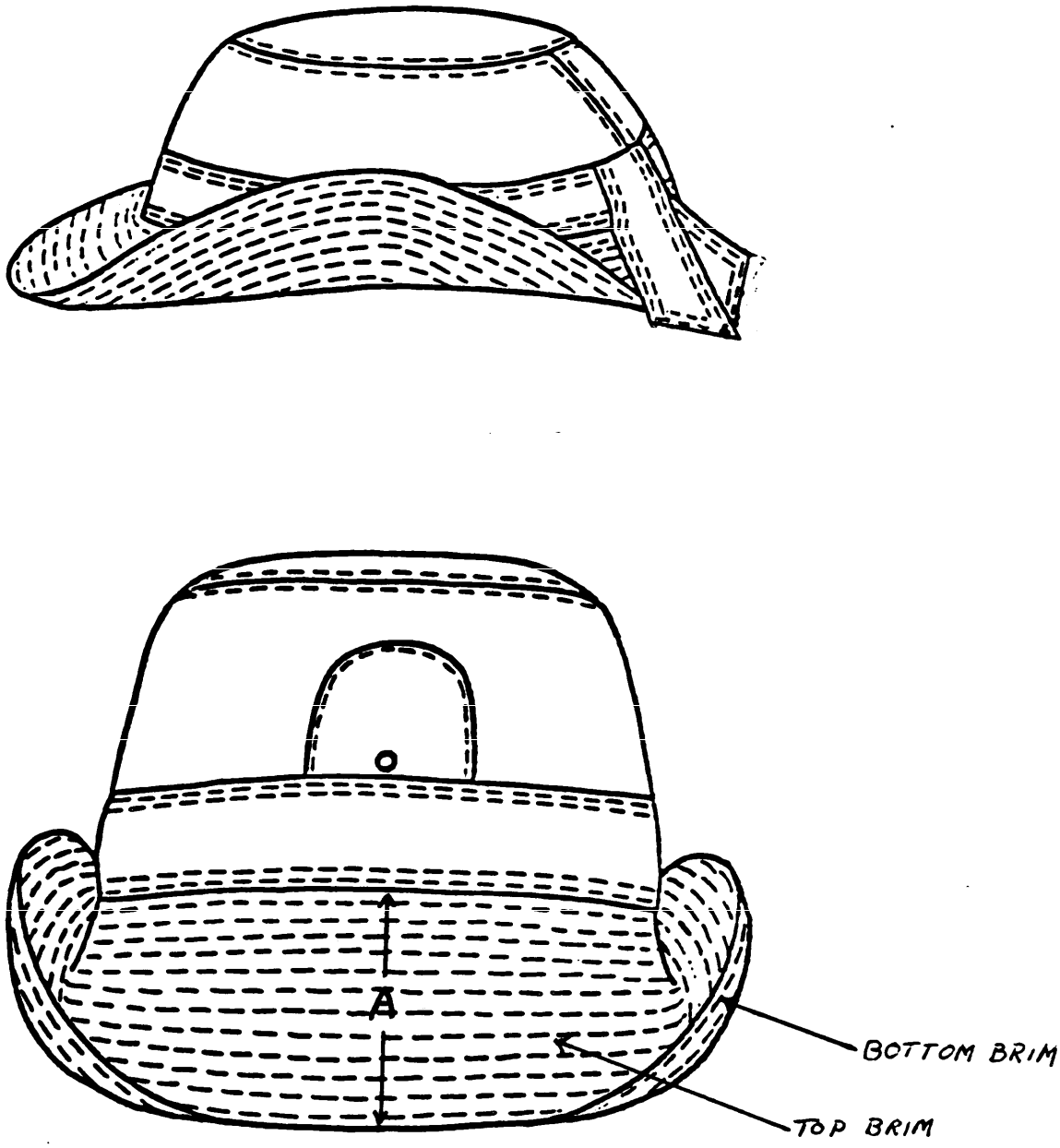
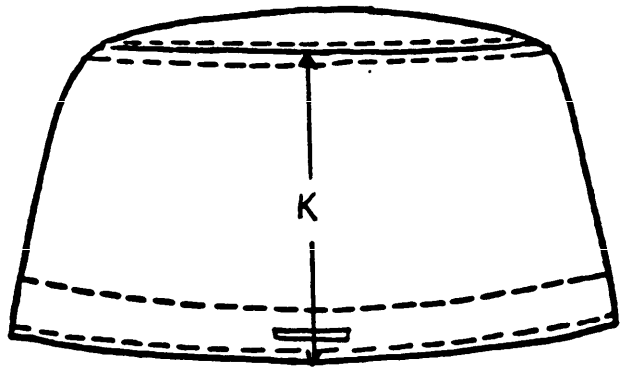
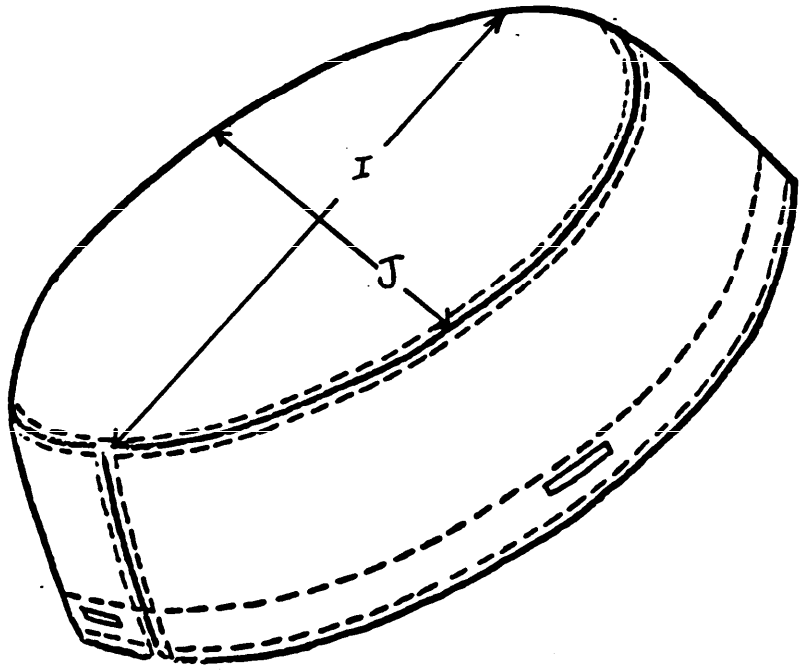


FIGURE 1 - HAT,SERVICE,WOMAN'S

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.



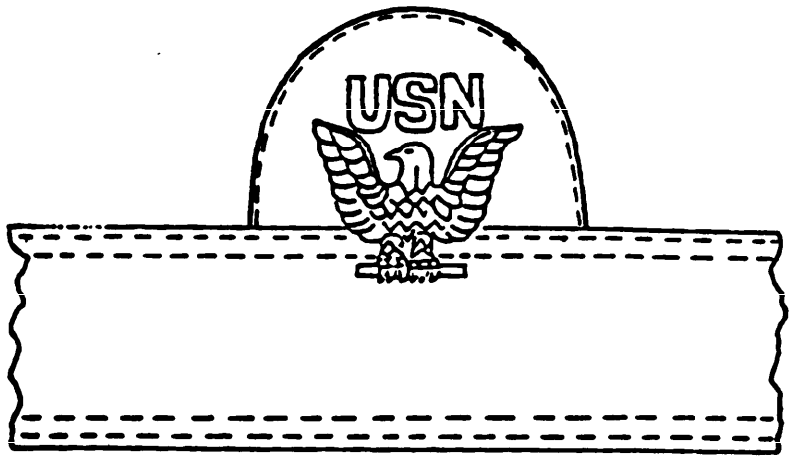
FRONT



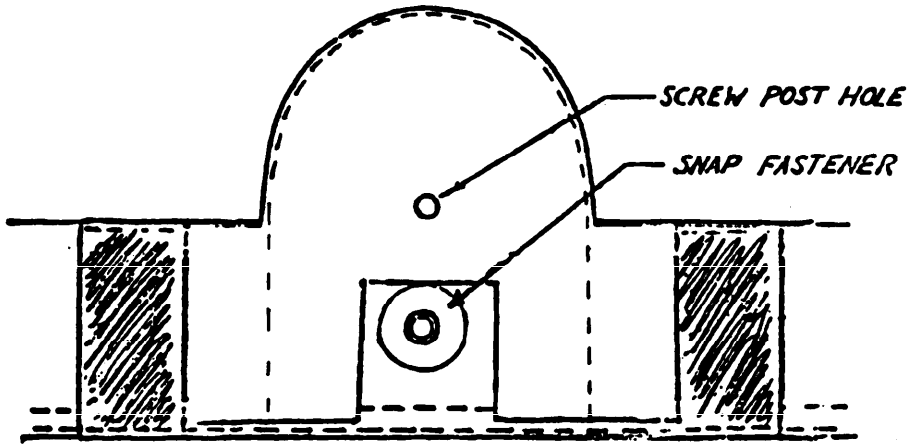
BACK AND SIDE

FIGURE 2 - HAT COVER, SERVICE WOMAN'S

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.



PLACEMENT OF INSIGNIA ON MOUNT AND BAND



PLACEMENT OF MOUNT ON BACK OF BAND

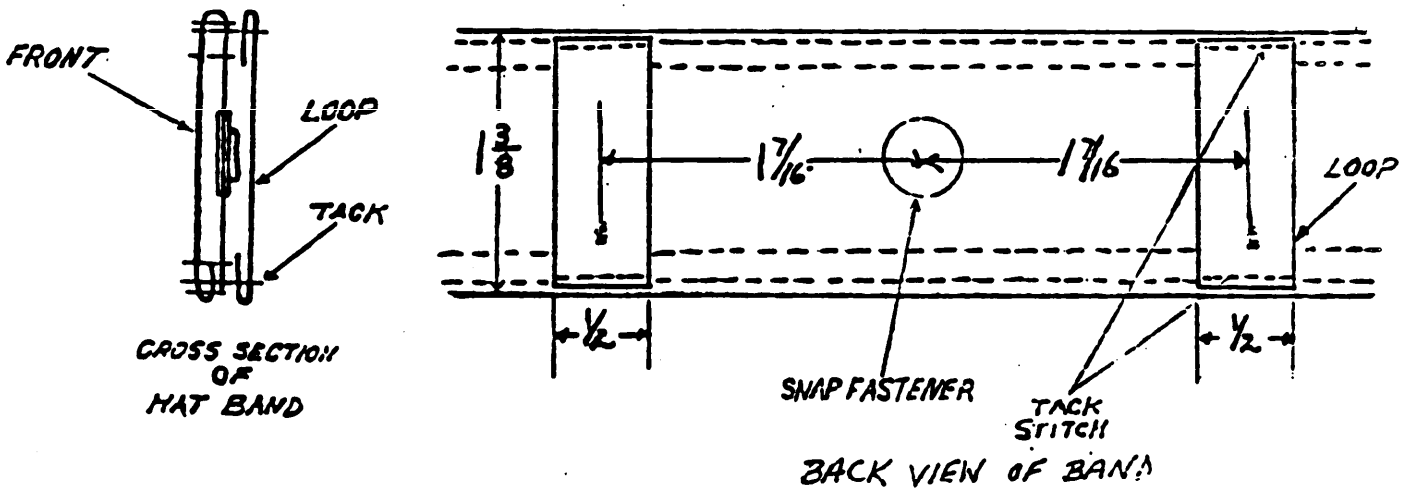


FIGURE 3 - HAT BAND and MOUNT

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

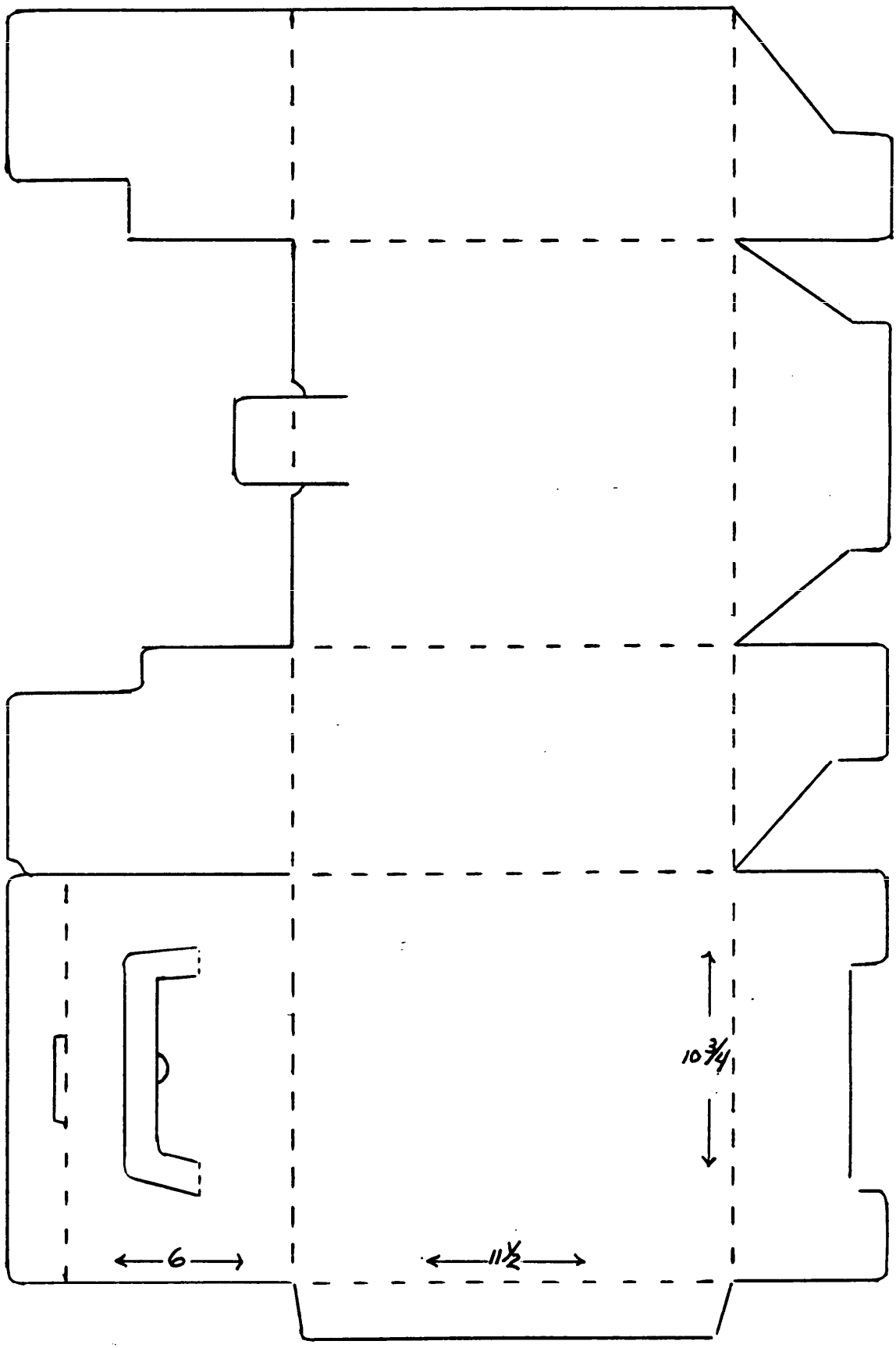


FIGURE 4 - HAT BOX

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
L-H-15505K(NU)	HAT, SERVICE, WOMAN'S

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 202610-18T06:17Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Officer in Charge
Navy Clothing & Textile Research Facility
21 Strathmore Road (Code 50)
Natick, MA 01760-5000

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:17Z
Check the source to verify that this is the current version before use.